

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 5

~ BookStream #002.02

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chapter 5 is called message passing and neurobiology what is your overview thought on chapter five okay uh I mean it's the kind of I don't know I have mixed feelings about this chapter because on one hand you see as far as I understand uh active inference although it originated as a quote-unquote a unified theory of the brain I don't think uh it's a neurobiological theory per se uh of course there can be some correlations between a neurobiological uh components or concepts with active inference active inference I mean Concepts but uh I mean it's not uh an essential premise of active inference Theory to provide a theory to provide a comprehensive theory about how the neurobiology of human brain or other organisms brain behave uh at the detailed and neural anatomical level but at the uh then again it's nice to have these kinds of empirical correlations uh between the findings of neurobiology and the active inference uh Theory uh but I I don't think it's uh one of active inference um Central assertions at least uh to my understanding well said very interesting framing well chapter five definitely takes a very specific system of Interest approach by highlighting one of the most studied areas also one of the most relevant areas which is mammalian neuroscience and the chapter is going to introduce a few different motifs in the nervous system and essentially build up towards figure 5.5 which is at the end of the chapter and 5.5 wires together three specific neural systems that the chapter is going to focus on work in that area from so Ollie said it very well active inference was built up to in chapter four here is another level or type of science with assertions or with representations or mappings to any specific system but this is the kind of modeling that has been built up and done by fristin power pizullo and others over the decades with a focus coming from a human neural Imaging laboratory setting a lot of focus and study and attention and funding and everything on the mammalian nervous system but claims about the nervous system have have uh are not the basis of what active inference claims or how it's derived but this is like an example case study in neurobiology connecting

back to some of the formalisms that we've just seen introduced in chapter four yeah and uh to add a minor Point uh to which you just said I think it's important to draw attention to the last sentence of the last paragraph of the first page it is important to draw a distinction between a principle I.E the minimization of free energy and a process theory about how this principle may be implemented in a certain kind of system so I I think this sentence here uh frames this chapter in relation to all the other technical chapters of this book so if every other chapter is about developing or at least up to now was about developing um the principled formalism of active inference now chapter 5 provides a kind of preliminary sketch for the process theory of active inference which is uh obviously far from an extensive Theory it's just a single chapter but then again it can provide some important uh signpost signpost for anyone who wants to further investigate this area awesome free energy principle Bayesian mechanics all things in that area are on this principle not responsive to empirical data and then the process theory is about how the principle is implemented so the specific generative models that are made and how well they map or how well they do in a portfolio of models that can have very different goals and assumptions and all of this but the process Theory implementation lets us develop hypotheses that are answerable to empirical data like what is the kind of information or relationship between photons hitting the retina and changes in activity in neural systems and that's an informational question or can be abstracted in a way to an informational question that it turns out does have empirical support and results in unique explanations and predictions that doesn't mean that it always results but a lot of citations are provided here that's what we can explore in chapter the last paragraph of the first section describes that they're going to look at the three different neural systems okay section 5.2 micro circuits and messages what do you think Ali uh all right so uh I mean this chapter begins uh from how a message passing happens in neurobiological terms um and compare it to the way active inference frames this message passing mechanism and specifically if we look at uh figure 5.1 and compare this figure to the ones we've seen before in chapters uh one through four I think it was in chapter four we can see some clear parallels between how this kind of cortical message passing happens in the brain versus how it is framed in active inference literature and as we can see it's uh clearly inspired by the neurobiology of the brain but then it's important to keep in mind that it's not a direct one-to-one mapping between these two models this is just um a kind of I don't know uh an interesting or Illuminating uh if you like parallel to keep in mind to somehow be a bit more confident about the viability of the theory we want to use for message passing and active inference uh which is to say it's not some haphazard Theory that's just been developed for

practical reasons it has some basis in neurobiology although uh it's not necessarily fully congruent with every detail of neurobiological neurobiology great the specific example is going to involve this one region of mammalian cortex tissue that has these six layers and there's a ton of neurobiology the big takeaway for figure 5.1 is that it's possible to graphically lay out nodes and variables and find some empirical correspondences again some unique explanations and predictions in certain cases and that's one kind of modeling where it's really trying to understand and improve the ability to do correlation and intervention and counterfactual causal type analysis with the real system of Interest or in a more pedagogical setting or in a research setting or an industrial setting you might sweep across large families of structures of models and there's no need to be grounded to any biological structure at all so this is just describing the specific neuroanatomical research that really arose out of the Imaging work at UCL and the SPM package so that's where a lot of this comes from 5.2 yeah good and sorry just as a side note I think watching uh one of Thomas parr's lectures on neurobiology of active inference which is available in YouTube would really help to understand the materials of this chapter better so I highly recommend watching that one thanks figure 5.2 gives a re-rendering of a kind of classical View of a hierarchical predictive coding system works so here abstracting a layer from the tissue six layer to just two layers here computational layers now and then showing how there's hierarchical communication within a layer but also uh well there's they're signaling within a layer and there's a hierarchy uh in Bayesian modeling with variables that are higher order predictions about other variables and that's the basis of the predictive coding architecture so 5.2 looks at some ways that the something that resonates with the cerebral cortical architecture enables what might computationally look like or have some really strong and explanatory values in actually relating to computationally a hierarchical Bayesian model which could do various General tasks all right 5.3 is motor commands leaving the prefrontal cortex going down to the butterfly looking cross section here what is 5.3 okay so uh 5.3 moves to the other half of active inference framework which is uh I mean how how it can model uh the decision making and uh ultimately uh the movement of the agent uh in order to minimize the expected free energy as opposed to variation on free energy that we saw in perceptual uh half of active inference so it can provide a kind of uh correlation or analogy between the structural neural Anatomy uh particularly related to uh I mean the motor commands and uh how it can relate to active inference in particularly The Continuous time active inference so we can see that uh for I mean for the external uh uh event or I'm sorry for the external State we can take for example the proprioceptive afferent and then uh this

proprioceptive afferent acts as a kind of why or the continuous time active inference which needs to be uh um I mean processed in in a way to optimize the expected free energy and how it relates to both attention and precision uh we will see a bit more detail about uh those uh those terms and the relation between them in chapter eight but I think here section 5.3 provides a good summary about the general uh paths through the motor command systems of neurobiology great let's say while the previous case study focused on how the connectivity within in between the cortical columns could have a computational relationship with a Bayesian hierarchical predictive coding architecture the argument of this second case study is that a continuous input continuous output kind of set point seeking reflexive motor Behavior with a moving set point with a descending moving set point enabling motion by changing ultimately the set point and enabling uh variation in the strategies to reach that set point through different mechanisms this is also describable in a compatible way that's a shorter section now section 5.4 subcortical structures what would you say about this section okay so subcortical structures are very important in the the decision making and and I mean the planning of the agents so uh obviously here we need to uh we need another kind of analogy between uh the that these plannings and decision makings happen uh neuroanatomically with the way uh that uh that it's framed in active inference but again uh we can see it's uh clearly based on uh I mean at least some of the important elements we've seen from uh the previous chapters uh so for example we we saw how policy is described or how how it relates to outcomes and and preference and so on uh we can see uh those elements are directly inspired by a neuroanatomical structures so I guess that's uh at least in my opinion this section here 5.4 uh seems a bit more uh sketchy in the meaning that it doesn't go into quite uh in the extensive details about how how how those structures can be um can be compared about for anyone who wants to uh further investigate these topics there are some useful references put on here and Pages 93 and 94. so yeah thanks yeah it's really abbreviated and overviewed but we get an interlude from table 5.1 with putative roles of neurotransmitters so same perspective that we took before on neural anatomical functionalism here directly translates to neurotransmitter reductionism or essentialism or something like that so certainly all neurotransmitters and molecules they play variable roles in different settings and this is the knee in scruffy manifold all over again one person might say well we we need a theory for every acetylcholine molecule in the world they're all in a unique context and someone else says all neurotransmitters are described by one parameter in this model I'm getting value from it so to me that's an account and somewhere in between is the work in this space which is making an

attempt to have a principled and falsifiable approach to model the computational aspects of specific regions and contexts and settings and so acetylcholine noradrenaline dopamine and serotonin are given a little mini review here and so it's not an exhaustive or an exclusive claim it's kind of a provocation from computational and molecular neuroscience and people can look into the papers and also ones that probably have been published since 5.6 goes to continuous and discrete hierarchies which is graphically overviewed in figure 5.5 so what would you say about this uh yeah one interesting thing about this section is the observation that our lower level engagement with the environment uh it can be most successfully characterized with continuous time formulations but as we go up uh and uh I mean on the level of uh cognitive Concepts or at the level of cognitive hierarchies and we come to Concepts such as I don't know decisions or even beliefs and so on we can reach the area that the discrete time situations uh would probably be more efficient to uh to characterize the behavior of the agent so this multi-scale um a multi-scale structure of active inference modeling is quite evident in the way that our message passing happens in our brain in terms of our lower level data processing up into consolidating the higher level cognitive Concepts and ontologies awesome thank you to me figure 5.5 demonstrates the kind of whole of body approach that you could imagine there's so many organs and systems and phenomena for which there aren't specific generative models so little can be said about situations where no generative model has been articulated and here's one where it has so it gives you also it's kind of like reading a drosophila melanogaster review paper relatively it's like this is how much work it takes to get to this state of knowledge in an insect so then in another insect do we know less about that insect empirically and genetically so consider this to be what's known to be a lot however also about one of the most sophisticated or specific cognitive systems at least we know so there's that additional kind of like self-reflexive aspect to this chapter that is not a Cornerstone of active inference but here it's just presented in a synthetic case study anything else you want to say about five uh nothing particular comes to mind all right